

UNIVERSITY OF THE PHILIPPINES
Tacloban College
Division of Natural Sciences and Mathematics
Bachelor of Science in Computer Science

CMSC 12 — Fundamentals of Programming 2
Second Semester AY 2025-2026

MACHINE PROBLEM

Pokerots: UP Tacloban Edition
Technical Report

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A. Rationale and Background

This project entails the implementation of a game modeled after classic Pokémon emulator games, but with a unique alteration. Instead of utilizing established Pokémon characters, the game will feature "Italian Brainrot" characters. These AI-generated fictional characters gained significant popularity through social media platforms such as TikTok, Instagram Reels, and YouTube Shorts.

Furthermore, the game will adopt the UP Tacloban Campus as the central "map" or environment. Key mechanics from the traditional Pokémon games will be incorporated, including NPC serving as opponents, and the core functionality of catching, owning, and engaging in battles with other *Pokerots*.

B. Objectives

The objectives are as follows:

- To apply concepts of encapsulation in handling data to prevent unnecessary access
- To apply multithreading in simulation of characters and other game concepts such as sound and dialogues.
- To implement Inheritance and Polymorphism for different entities and PokéRot types.
- To utilize arrays for tile-based map generation and exploration logic.
- To apply File I/O for saving and loading player progress and that of captured PokéRots
- To design a Graphical User Interface (GUI) using Swing or other Java features for immersive gameplay.
- To implement Event Handling for player movement and interaction with the environment.
- To develop a game loop for real-time rendering and state updates.

C. Tools and Technologies

- Java 25
- Piskel
- LibreSprite
- Tiled Map Editor
- Bandlabs
- Beepbox

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D. Functional Requirements (Gameplay Dynamics)

This section describes the functional requirements, specifically the gameplay dynamics of the program, as well as additional features like progress saving, inventory, etc:

Movement and Exploration: The player will navigate the UP Tacloban Campus map using tile-based, 4-directional movement. Throughout the game the player will encounter NPC's that will battle with the player. The system will handle collision detection against environmental obstacles (e.g., campus buildings, fences, trees). Specific tile zones will trigger random encounters with wild "Italian Brainrot" characters which will trigger battles.

Combat and Battle System: Battles will be turn-based, and the calculations will rely on entity stats (health, attack damage).

Catching and Storing PokeRots: The player can catch wild PokeRots using specific in-game items such as the "Malunggay Pandesal" in which the catch-rate depends on the health bar of the enemy PokeRot. Those PokeRots will be stored in the player's bag which can be used to fight other PokeRots.

Progression System: PokeRots gain experience points (XP) from battles, allowing them to level up and increase their base stats. Players progress through the game by locating and defeating NPC's scattered across the campus map.

Win/Loss Conditions: Successfully defeating all NPC's of the UP Tacloban Campus results in you "completing" the game, although the game does not have a clear ending since that feature is still a work in progress. If all PokeRots in the player's active party lose their HP, the player will lose all of his PokeRot's except his starting PokeRot.

Data Persistence: The game will utilize File I/O to save and load the player's current location, inventory, party stats, and overall progress.

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E. Methodology

The architecture will rely heavily on core Object-Oriented Programming principles to ensure the codebase remains modular, scalable, and easy to debug during the testing phase. The visual component of the game will feature a Graphical User Interface to create an immersive gameplay experience.

Game Asset Creation:

- We will primarily design and generate our 2D game assets from scratch, utilizing online resources only as source material for supplementary content.
- The graphics development will be made using pixel-art software, specifically Piskel, and using LibreSprite and Tiled Map Editor to make our map.

Class Hierarchy and OOP Design:

- By overriding standard methods we can ensure that an `interact()` method behaves differently depending on which NPC the player interacts with, in terms of dialogue and triggering a battle.
- Statistics for PokeRots (Health, Experience, Attacks) and the player's inventory will be encapsulated using private variables, accessible only via getter and setter methods.

Game State Management

To manage the flow of the game cleanly, we will implement a Game State System using Enums. This will separate the game into distinct operational states, such as ROAMING (exploring the map), BATTLE (engaging in combat) which is divided into battling with NPCs and battling with Pokemons, and DIALOGUE (interacting with NPCs). This ensures that event handling logic does not overlap. For example, preventing the player from accidentally moving their character in the background while they are locked in a turn-based battle or reading text.

Game Loop and Multi-Threading

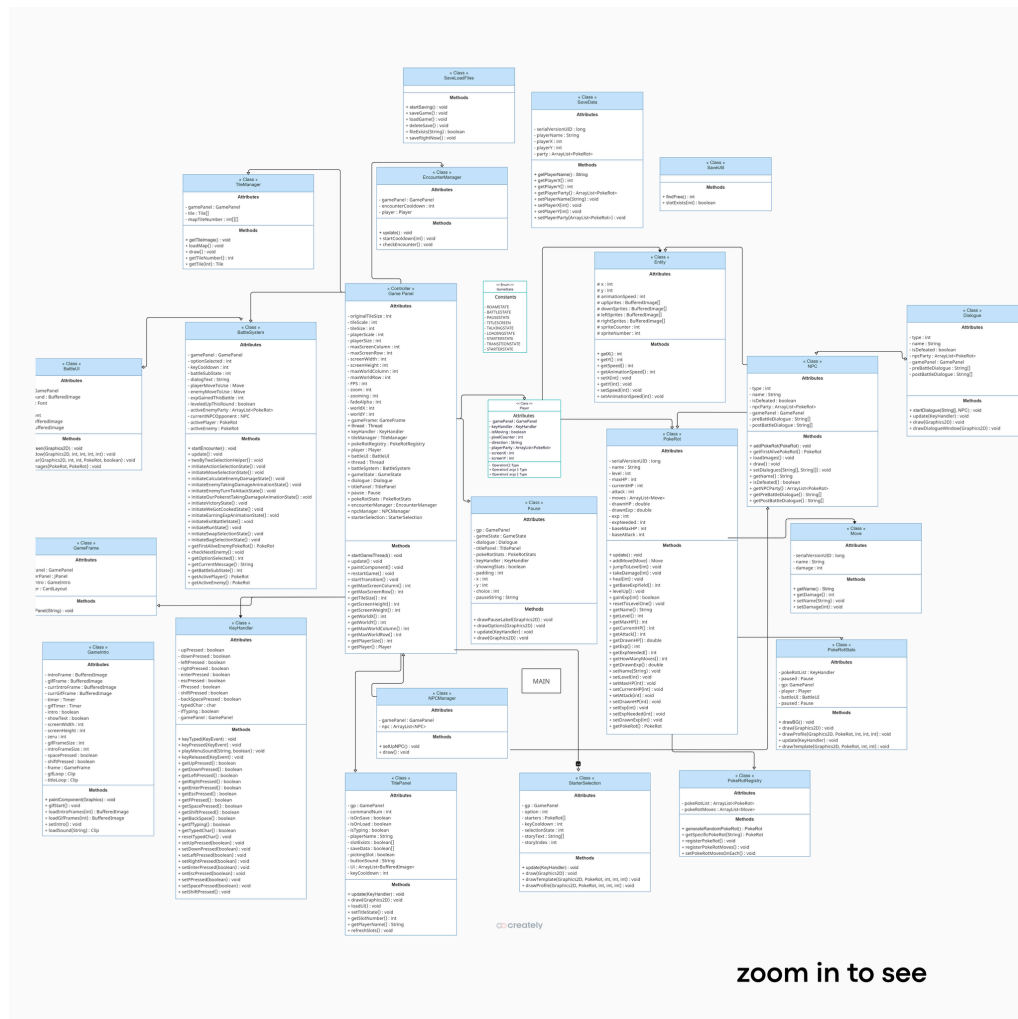
- We will develop a real-time game loop responsible for continuous rendering and state updates.
- Event Handling will be integrated into this loop to capture real-time keyboard inputs, translating them into player movement and environmental interactions.
- During battle events, multi-threading will allow two opposing Pokemons to interact as separate entities. This ensures that combat actions such as fighting, running, or catching can be processed concurrently without freezing the main GUI.

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Map Generation and Data Management

- The map will utilize a 2D array for tile-based map generation. The grid system will dictate movement paths, trigger wild encounters, and manage environmental collisions.
- For data persistence, the game will apply File I/O operations to handle the saving and loading of game progress. This system will securely store the player's current map location, the contents of their inventory, captured PokeRots, and party stats into a local file so progress is not lost upon program termination.

Class Diagram

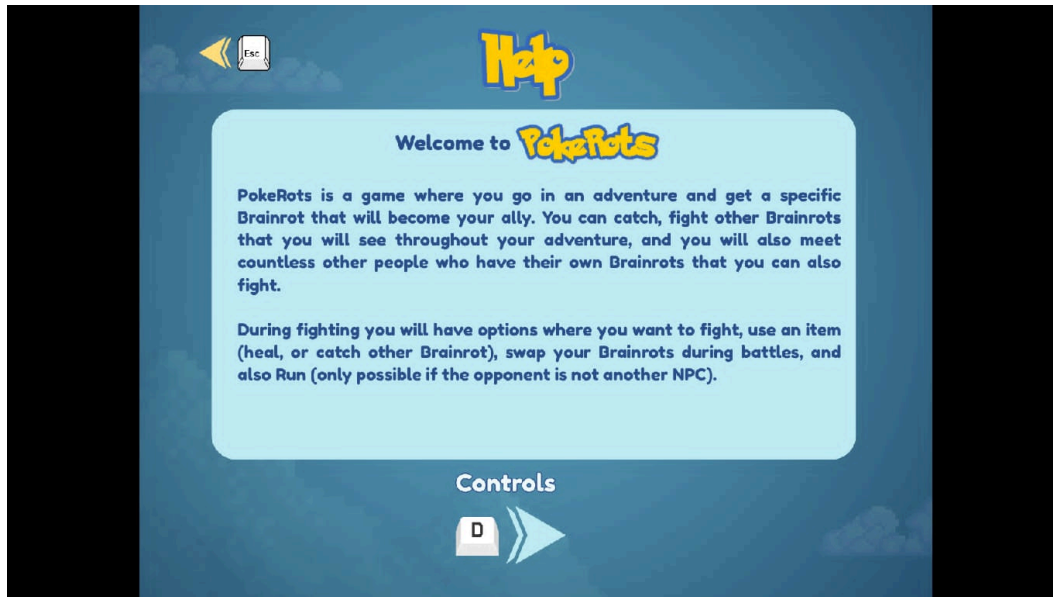


zoom in to see

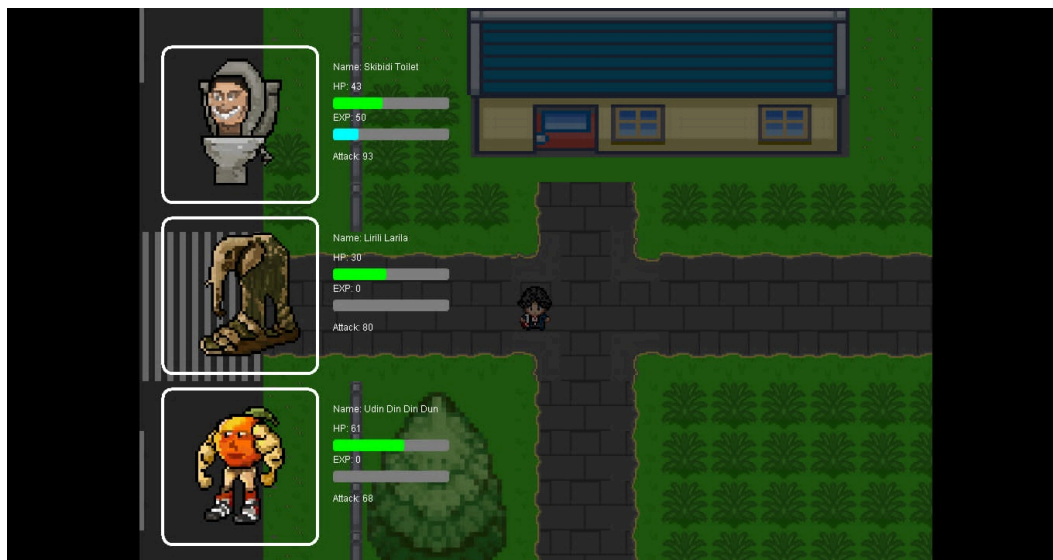
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F. Results

Tutorial/Help Panel:

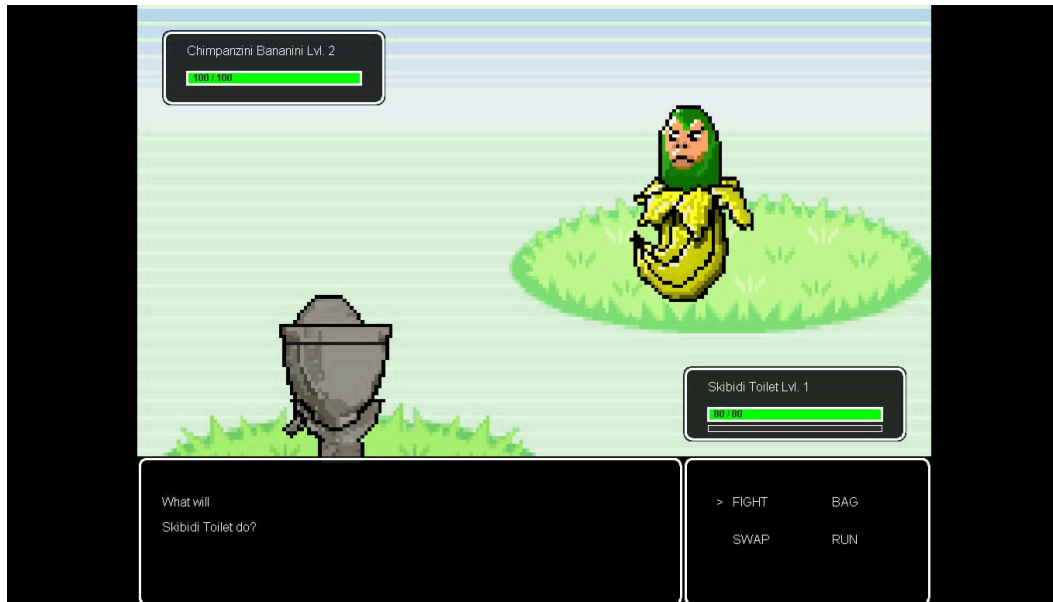


PokeRot Inventory Panel:

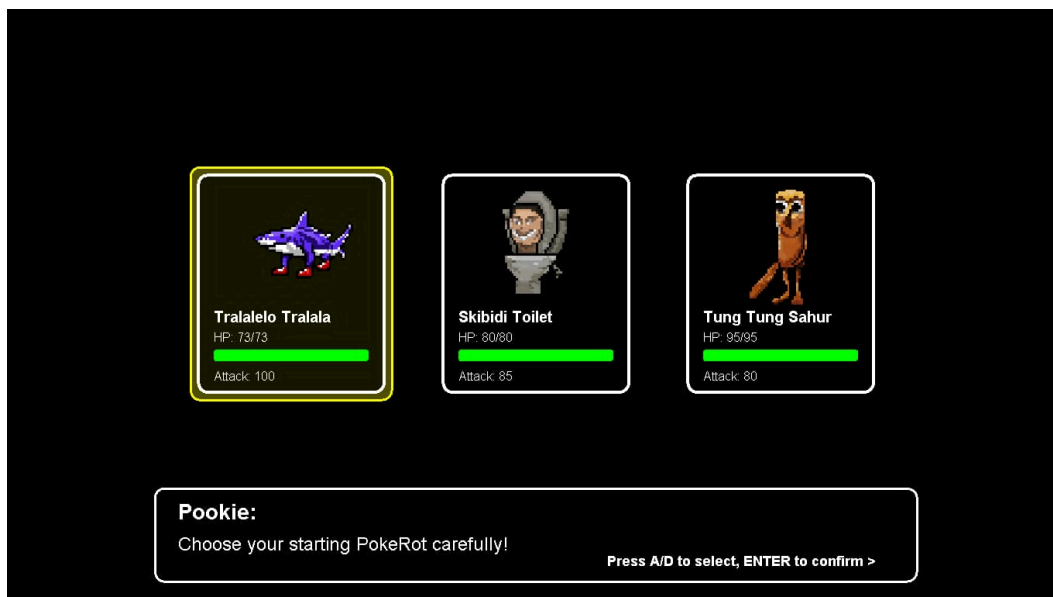


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Battle State Panel:

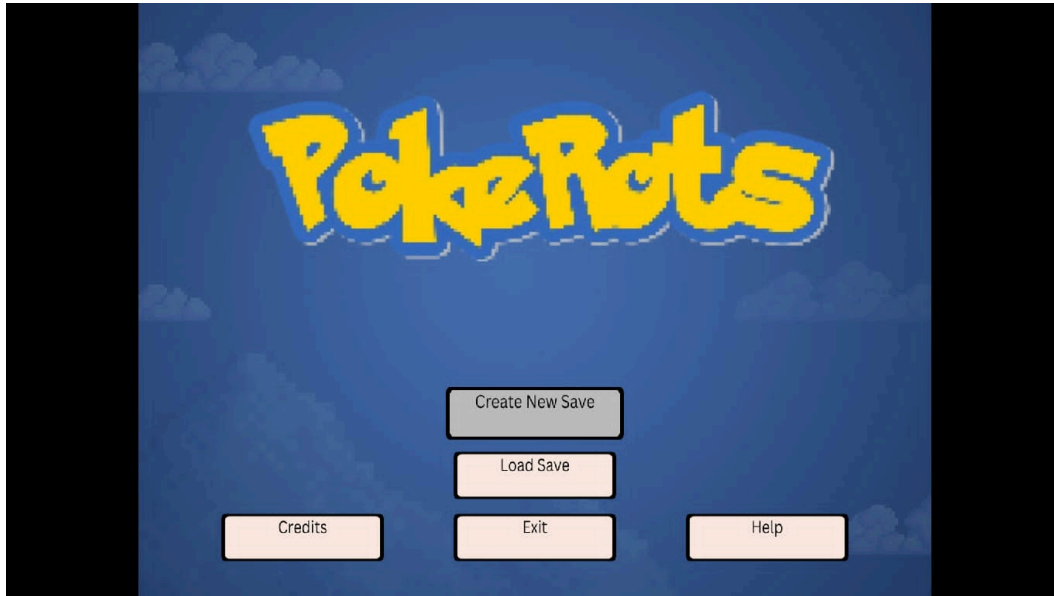


Choosing Starting PokeRot Panel:



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Home Screen Panel:



G. Problems/Bugs

List of bugs and problems for future devs:

- You will need to save your progress first before leaving the game, because if you play/open the game again and do not save your progress, your progress is lost since the game does not have an autosave feature.